

Independent Study Progress Report - David Chen

Original Prompt/Proposal

This research addresses the gap in generative AI's ability to realistically simulate texture deterioration in architectural facades over time. Current models predominantly focus on generating pristine, "new-looking" textures for architectural presentations, neglecting the critical aspect of how materials age due to environmental exposure. Unlike existing AI models that excel in simulating aging in human faces, architectural materials such as stone, brick, and metal are often rendered with overly simplistic or stereotypical depictions of wear, failing to capture the nuanced and varied patterns of deterioration.

The project aims to develop a generative AI framework capable of simulating realistic aging in facade materials, offering architects and designers a tool to visualize the long-term effects of environmental exposure on their designs. By leveraging datasets that document real-world aging patterns and integrating advanced tools like ComfyUI, this research seeks to refine generative models to produce aged textures that are both realistic and contextually relevant. This approach encourages architects to incorporate the factor of time into their design process, fostering a more holistic representation of their work.

Research Objectives

- **Identify limitations** in current generative models for simulating aged architectural textures.
- **Develop or source datasets** that accurately capture the diverse aging patterns of materials under different environmental conditions.
- **Refine diffusion models** using tools like ComfyUI to enhance their ability to simulate material weathering.
- **Test and validate** the framework by generating realistic aged textures and comparing them to real-world examples.

Current Progress and Directions

Below is a flow chart of how the overall workflow might look like integrating functionality of analysis, visualization, and finally suggestion. Each section will use specifically modules and libraries that are the most suitable and scalable for the intended purpose. Currently I have been focusing on mainly deploying llama 3 models to generate real-time feedback for captured materials on the screen. (Figure 1)

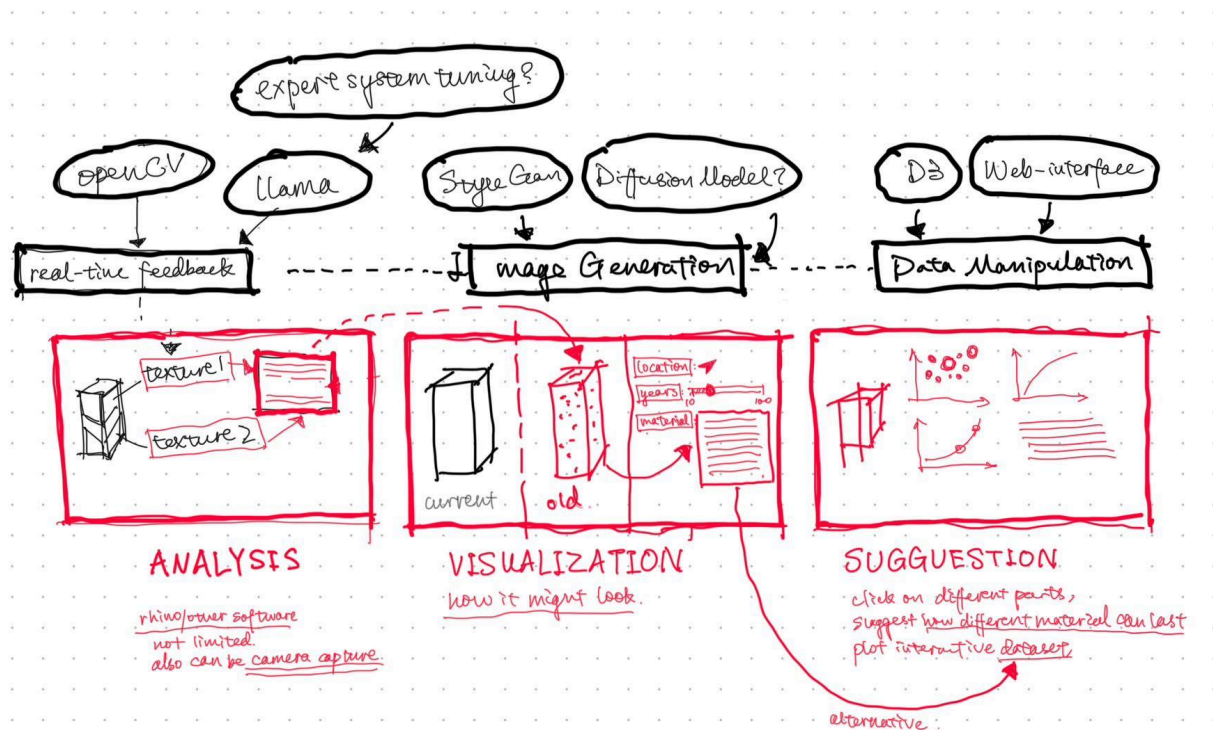


Figure 1: overall intended program flowchart

I have constructed the front-end and back-end scaffolding of the project, including a web-page running through svelte and a backend that uses llama 3 running locally to read captured screens and generate description of the material shown, trying to predict how many years have deteriorated for the texture. Below is the project framework involving appropriate tools used. (figure 2)

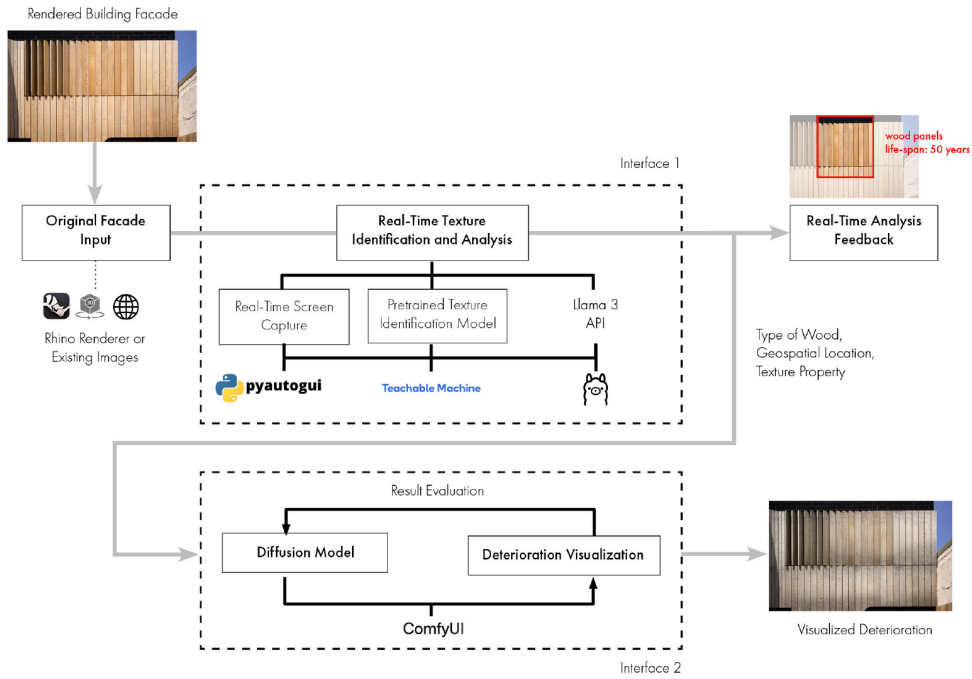
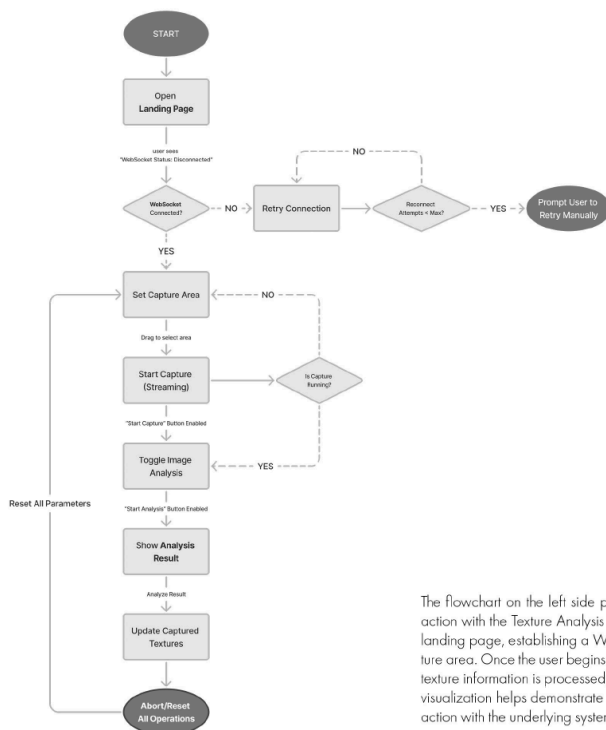


Figure 2 Analysis + Visualization flow chart



User Action Flow Chart

Figure 3 Interaction Flowchart

The flowchart above presents the overall user journey and interaction with the Texture Analysis Tool. The workflow starts with opening the landing page, establishing a WebSocket connection, and setting the capture area. Once the user begins capturing and analyzing images, real-time texture information is processed, displayed, and updated accordingly. This visualization helps demonstrate the tool's ease of use and its efficient interaction with the underlying system.

While still having unexpected bugs and algorithmic limitations, a running wireframe prototype screenshot is a separate window hovering above the rendering window of a design software, and captures the render output in real time to read, analyze and finally export a deteriorated version of the rendered texture. (Figure 4)

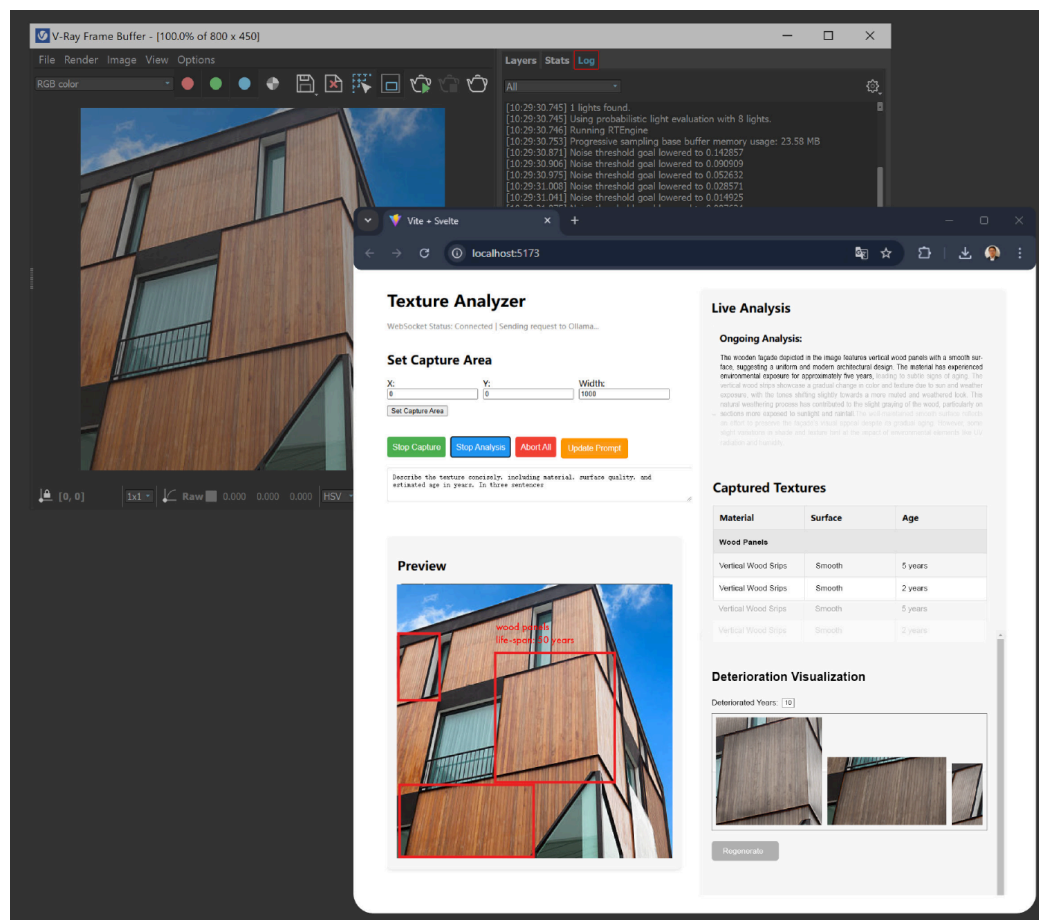


Figure 4 Web-interface Prototype

Reflections and Future Works

This semester's work revealed several challenges in the analysis methodology, particularly in deploying LLama 3 locally. The attempt to use LLama 3 for image recognition was largely unsuccessful due to the model's need for significantly greater computational resources. Since LLama 3 with image recognition capabilities requires a much larger model, running it locally is impractical. As a result, the next step will involve transitioning the system to Microsoft Azure, where cloud resources can properly support the deployment of LLama 3 for tasks like screen capture and analysis.

Additionally, while the web interface offers some benefits, it has proven to be less intuitive compared to a dedicated local application. Future efforts may explore refining the interface to balance accessibility with usability, possibly considering local application development as an alternative.

The system is currently limited to detecting a single texture, which restricts its broader applicability. To address this, integrating an image segmentation and classification module before generating descriptions with LLama 3 will be a necessary improvement. This addition could enable the system to handle multiple textures and provide richer outputs.

One of the most significant limitations encountered was the lack of tagged or labeled data for deteriorated textures. Despite efforts to generate procedural textures using tools like Blender and other applications, integrating scientifically plausible methodologies to predict and simulate material aging remains a substantial challenge. This underscores the need for alternative approaches to simulate and analyze texture deterioration in a scientifically robust manner.

Looking ahead, I plan to reevaluate the plausibility of creating visually and scientifically accurate simulations of material deterioration. This reevaluation will involve exploring alternative methods to achieve reliable analyzation and visualization functionalities. Overall, this semester's work served as a test drive to address these challenges both

technically and ideologically. The findings and resources gathered will provide a strong foundation for constructing more comprehensive frameworks in the future.

References and Resources

Workflow Tools and Frameworks

- ComfyUI for Flux ControlNet Depth and Canny Workflows:
<https://www.runcomfy.com/comfyui-workflows/comfyui-flux-controlnet-depth-and-canny>
- ControlNet GitHub Repository:
<https://github.com/lllyasviel/ControlNet>

Texture Analysis and Deterioration

- T2D2 Damage Detector Framework:
<https://t2d2.ai/damage-detector>
- Master's Thesis on Texture Analysis:
https://graphics.cs.ucy.ac.cy/files/1642536375_Thesis-final.pdf
- Synthesizing Time-Varying Weathered Textures:
<https://www.cs.tau.ac.il/~dcor/articles/2016/TW.pdf>
- Use of Weathered Surfaces in Building Simulation:
<https://www.sciencedirect.com/science/article/pii/S0360132317305243>

Machine Learning and Vision Models

- LLama 3.2 Vision-Instruct Model (Azure):
<https://ai.azure.com/explore/models/Llama-3.2-11B-Vision-Instruct/version/1/register/azureml-meta>

Architectural Data

- Historic American Engineering Record, HAER IL-17:
<https://tile.loc.gov/storage-services/master/pnp/habshaer/il/il0000/il0017/data/il0017data.pdf>